DATA ANALYSIS REPORT

Introduction

This report presents a comprehensive analysis of an e-commerce dataset using SQL to uncover meaningful business insights. The analysis covers key aspects such as customer behaviour, sales performance, product trends, seller performance, payments, deliveries, and customer reviews. The objective is to transform raw transactional data into actionable insights that can support data-driven business decisions and improve overall business performance.

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1. Database Exploration

Purpose: Understand the dataset.

Question 1 : How many records are in each table?

Output :

table_name	total_records
category_translation	71
sellers	3095
products	32951
order_reviews	99224
customers	99441
orders	99441
order_payments	103886
order_items	112650

Observation

The database contains 8 tables. The order_items table has the highest number of records (112,650), while category_translation has the fewest (71). The orders and customers tables each contain 99,441 records.

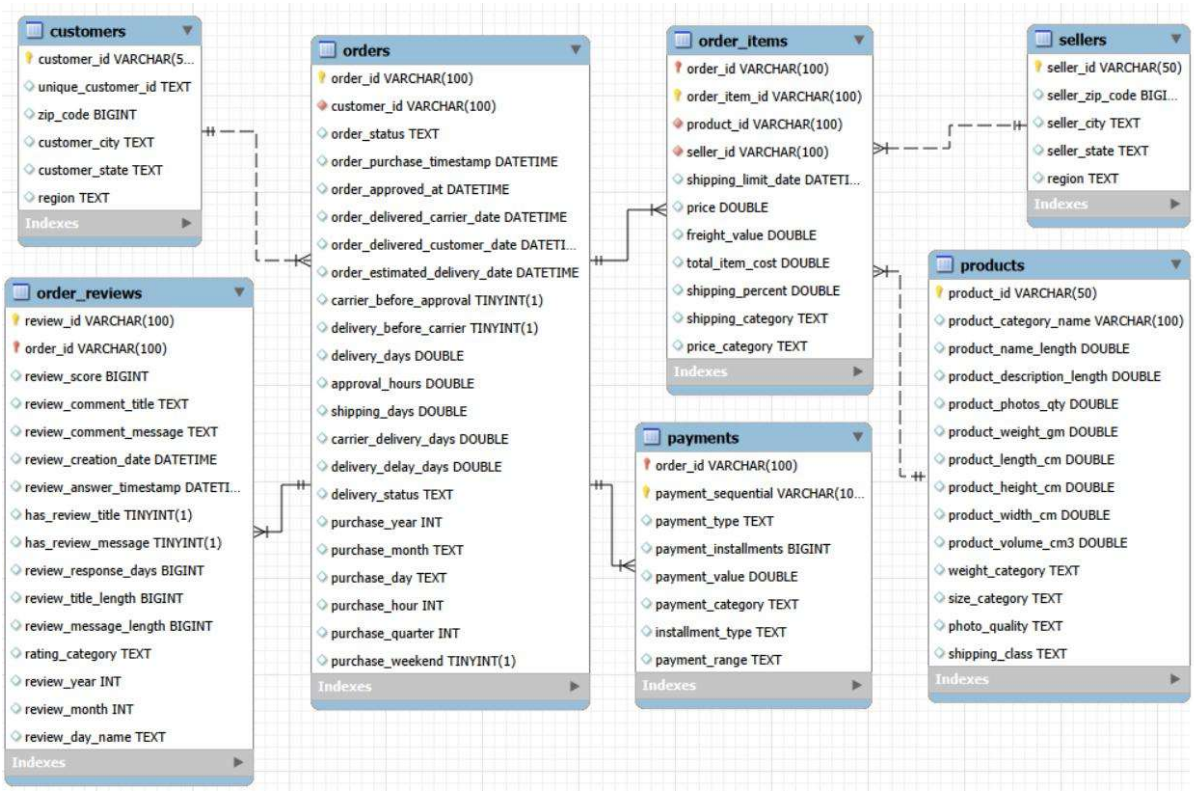
Key Insights

- More records in order_items than orders indicate that many orders contain multiple products.
- The higher count in order_payments suggests some orders involve multiple payment transactions.

- The large number of products and sellers reflects a diverse e-commerce marketplace.

Question 2 : What is the relationship between all tables?

Output :



Observation

The database follows a **relational schema** where the orders table acts as the central table, connecting customers, order items, payments, and reviews. Products and sellers are linked through the order_items table

Key Insights

- The schema is well-structured and normalized, reducing data redundancy.
- The central orders table enables efficient analysis of customer, product, payment, seller, and review data in a single workflow.

Question 3 : What is the date range of orders?

Output :

first_order_date	last_order_date
2016-09-04 21:15:19	2018-10-17 17:30:18

Observation

The dataset contains orders placed between **4 September 2016** and **17 October 2018**, covering a period of about **2 years**.

Key Insights

- The dataset provides sufficient historical data to analyse sales trends, customer behaviour, and seasonal patterns over multiple years.

Question 4 : How many unique customers, sellers, products, and categories exist?

Output :

Metric	Count
Unique_Customers	99441
Unique_Sellers	3095
Unique_Products	32951

Observation

The dataset contains 99,441 unique customers, 3,095 unique sellers, and 32,951 unique products.

Key Insights

- The high number of customers and products indicates a large and diverse e-commerce marketplace with a broad product catalog.

Question 5 : What are the distributions of order status, payment types, and review scores?

Output :

order_status	total_orders
delivered	96478
shipped	1107
canceled	625
unavailable	609
invoiced	314
processing	301
created	5
approved	2

payment_type	total_payments
credit_card	76795
boleto	19784
voucher	5775
debit_card	1529
not_defined	3

review_score	total_reviews
1	11424
2	3151
3	8179
4	19142
5	57328

Observation

Most orders were successfully delivered (96,478). Credit cards (76,795) were the most commonly used payment method, and 5-star reviews (57,328) were the most frequent customer rating.

Key Insights

- The platform maintains a high order completion rate, with delivered orders dominating the dataset.
- Customers prefer credit card payments over other payment methods.
- The large share of 5-star reviews reflects a generally positive customer experience.

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2. Sales & Product Performance

Purpose: Analyse revenue and product performance.

Question 6 : What is the total revenue?

Output :

total_revenue
16008872.12

Observation

The total revenue generated during the analysis period is 1,600,872.12.

Key Insights

- The platform generated over 1.6 million in revenue, indicating strong overall sales performance during the observed period.

Question 7 : What is the monthly revenue trend?

Output :

month	total_revenue
September	732454.23
April	1578573.51
January	1253492.22
August	1696821.64
February	1284371.35
May	1746900.97
December	878421.1
July	1658923.67
March	1609515.72
November	1194882.8
October	839358.03
June	1535156.88

Observation

Revenue varies across the months. May (1,746,900.97) recorded the highest revenue, while September (732,454.23) generated the lowest.

Key Insights

- Monthly revenue shows noticeable fluctuations, indicating seasonal changes in customer demand and sales performance.

Question 8 : What is the Average Order Value (AOV)?

Output :

average_order_value
160.99

Observation

The Average Order Value (AOV) is 160.99.

Key Insights

- On average, each order contributes 160.99 in revenue, providing a useful benchmark for evaluating customer spending and sales performance.

Question 9 : Which product categories generate the highest revenue?

Output :

product_category_name	total_revenue
beleza_saude	1258681.34
relogios_presentes	1205005.68
cama_mesa_banho	1036988.68
esporte_lazer	988048.97
informatica_acessorios	911954.32

Observation

The Beleza_Saude category generated the highest revenue (1,258,681.34), followed by Relogios_Presentes and Cama_Mesa_Banho.

Key Insights

- A few product categories contribute a significant share of total revenue, highlighting them as key drivers of overall sales.

Question 10 : Which products generate the highest revenue?

Output :

product_id	total_revenue
bb50f2e236e5eea0100680137654686c	63885
6cdd53843498f92890544667809f1595	54730.2
d6160fb7873f184099d9bc95e30376af	48899.34
d1c427060a0f73f6b889a5c7c61f2ac4	47214.51
99a4788cb24856965c36a24e339b6058	43025.56
3dd2a17168ec895c781a9191c1e95ad7	41082.6
25c38557cf793876c5abdd5931f922db	38907.32
5f504b3a1c75b73d6151be81eb05bdc9	37733.9
53b36df67ebb7c41585e8d54d6772e08	37683.42
aca2eb7d00ea1a7b8ebd4e68314663af	37608.9

Observation

The highest-performing product generated 63,885 in revenue, followed by several other products with revenues ranging from 37,609 to 54,730.

Key Insights

- A small group of products contributes significantly to total revenue, making them key products for sales and inventory planning.

Question 11 : Which states contribute the most revenue?

Output :

customer_state	total_revenue
SP	5998226.96
RJ	2144379.69
MG	1872257.26
RS	890898.54
PR	811156.38
SC	623086.43
BA	616645.82
DF	355141.08
GO	350092.31
ES	325967.55

Observation

São Paulo (SP) generated the highest revenue (5,998,226.96), followed by Rio de Janeiro (RJ) and Minas Gerais (MG).

Key Insights

- Revenue is concentrated in a few states, with SP being the primary contributor to overall sales.

Question 12 : Which cities contribute the most revenue?

Output :

customer_city	total_revenue
Sao Paulo	2203373.09
Rio De Janeiro	1161927.36
Belo Horizonte	421765.12
Brasilia	354216.78
Curitiba	247392.48
Porto Alegre	224731.42
Salvador	218071.5
Campinas	216248.43
Guarulhos	165121.99
Niteroi	139996.99

Observation

São Paulo generated the highest revenue (2,203,373.09), followed by Rio de Janeiro and Belo Horizonte.

Key Insights

- Revenue is concentrated in major metropolitan cities, with São Paulo being the largest contributor to overall sales.

Question 13 : Which price category generates the highest revenue?

Output :

price_category	revenue
Standard	7700330.73
Premium	6342420.65
Luxury	3486262.97
Budget	2779120.36

Observation

The Standard price category generated the highest revenue (7,700,330.73), followed by Premium, Luxury, and Budget.

Key Insights

- Standard-priced products are the primary revenue drivers, indicating that customers purchase mid-priced products more frequently than premium or budget options.

Question 14 : Which products are sold the most?

Output :

product_id	total_units_sold
aca2eb7d00ea1a7b8ebd4e68314663af	527
99a4788cb24856965c36a24e339b6058	488
422879e10f46682990de24d770e7f83d	484
389d119b48cf3043d311335e499d9c6b	392
368c6c730842d78016ad823897a372db	388
53759a2ecddad2bb87a079a1f1519f73	373
d1c427060a0f73f6b889a5c7c61f2ac4	343
53b36df67ebb7c41585e8d54d6772e08	323
154e7e31ebfa092203795c972e5804a6	281
3dd2a17168ec895c781a9191c1e95ad7	274

Observation

The top-selling product recorded 527 units sold, while the remaining top products sold between 274 and 488 units.

Key Insights

- A small number of products account for the highest sales volume, making them important for inventory management and demand planning.

Question 15 : Which product categories sell the most?

Output :

product_category_name	total_units_sold
cama_mesa_banho	11115
beleza_saude	9670
esporte_lazer	8641
moveis_decoracao	8334
informatica_acessorios	7827
utilidades_domesticas	6964
relogios_presentes	5991
telefonia	4545
ferramentas_jardim	4347
automotivo	4235

Observation

The Cama_Mesa_Banho category sold the highest number of units (11,115), followed by Beleza_Saude and Esporte_Lazer.

Key Insights

- Home, health, and sports-related categories have the highest sales volume, indicating strong customer demand in these segments**.**

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3. Customer Analytics

Purpose: Understand customer behaviour.

Question 16 : How many customers are new vs repeat customers?

Output :

customer_type	customer_count
New Customer	93099
Repeat Customer	2997

Observation

The dataset contains 93,099 new customers and 2,997 repeat customers.

Key Insights

- Most customers made only one purchase, indicating a low repeat purchase rate and an opportunity to improve customer retention.

Question 17 : Who are the top 10 customers by revenue?

Output :

unique_customer_id	total_revenue
0a0a92112bd4c708ca5fde585afaa872	13664.08
46450c74a0d8c5ca9395da1daac6c120	9553.02
da122df9eeddfedc1dc1f5349a1a690c	7571.63
763c8b1c9c68a0229c42c9fc6f662b93	7274.88
dc4802a71eae9be1dd28f5d788ceb526	6929.31
459bef486812aa25204be022145caa62	6922.21
ff4159b92c40ebe40454e3e6a7c35ed6	6726.66
4007669dec559734d6f53e029e360987	6081.54
5d0a2980b292d049061542014e8960bf	4809.44
eebb5dda148d3893cdaf5b5ca3040ccb	4764.34

Observation

The highest-spending customer generated 13,664.08 in revenue, while the other top 10 customers generated between 4,764.34 and 9,553.02.

Key Insights

- A small group of high-value customers contributes significantly to total revenue, making them ideal candidates for loyalty and retention programs.

Question 18 : What is the Customer Lifetime Value (CLV)?

Output :

Observation

The highest-spending customer generated 13,664.08 in revenue, while the other top customers contributed between 4,764.34 and 9,553.02.

Key Insights

- A small group of high-value customers contributes significantly to revenue, making them important for customer retention and loyalty programs.

Question 19 : Which region has the highest customer base?

Output :

region	customer_base
Southeast	65900
South	13690
Northeast	9140
Central-West	5595
North	1794

Observation

The Southeast region has the highest number of unique customers (65,900), followed by the South (13,690) and Northeast (9,140).

Key Insights

- The Southeast is the largest customer market, indicating the strongest customer presence and growth potential.

Question 20 : What is the average purchase frequency?

Output :

avg_purchase_frequency
1.03

Observation

The average purchase frequency is 1.03 orders per customer.

Key Insights

- Most customers place only one order, indicating limited repeat purchasing and an opportunity to improve customer retention.

Question 21 : How many customers have become inactive?

Output :

inactive_customers
67580

Observation

The dataset contains 67,580 inactive customers.

Key Insights

- A large inactive customer base indicates an opportunity to increase repeat purchases through targeted retention and re-engagement campaigns.

Question 22 : Which states generate the most orders?

Output :

state	order_count
SP	41746
RJ	12852
MG	11635
RS	5466
PR	5045
SC	3637
BA	3380
DF	2140
ES	2033
GO	2020
PE	1652
CE	1336
PA	975
MT	907
MA	747

Observation

São Paulo (SP) recorded the highest number of orders (41,746), followed by Rio de Janeiro (RJ) and Minas Gerais (MG).

Key Insights

- Order volume is heavily concentrated in SP, making it the largest market and a key region for sales and operational planning.

Question 23 : Which city have the highest customer concentration?

Output :

city	customer_count
Sao Paulo	14984

Observation

São Paulo has the highest customer count, with 14,984 customers.

Key Insights

- São Paulo is the largest customer hub, making it a key market for customer acquisition and retention strategies.

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4. Seller and logistic analysis

Purpose: Evaluate seller and delivery performance.

Question 24 : Who are the top sellers by revenue?

Output :

seller_id	total_revenue
7c67e1448b00f6e969d365cea6b010ab	507166.91
1025f0e2d44d7041d6cf58b6550e0bfa	308222.04
4a3ca9315b744ce9f8e9374361493884	301245.27
1f50f920176fa81dab994f9023523100	290253.42
53243585a1d6dc2643021fd1853d8905	284903.08
da8622b14eb17ae2831f4ac5b9dab84a	272219.32
4869f7a5dfa277a7dca6462dcf3b52b2	264166.12
955fee9216a65b617aa5c0531780ce60	236322.3
fa1c13f2614d7b5c4749cbc52fecda94	206513.23
7e93a43ef30c4f03f38b393420bc753a	185134.21

Observation

The top-performing seller generated 507,166.91 in revenue, while the remaining top sellers generated between 185,134.21 and 308,222.04.

Key Insights

- A small group of sellers contributes a significant share of total revenue, making them key partners for business growth and performance.

Question 25 : Who processes the highest number of orders?

Output :

seller_id	TotalOrders
6560211a19b47992c3666cc44a7e94c0	1854
4a3ca9315b744ce9f8e9374361493884	1806
cc419e0650a3c5ba77189a1882b7556a	1706
1f50f920176fa81dab994f9023523100	1404
da8622b14eb17ae2831f4ac5b9dab84a	1314
955fee9216a65b617aa5c0531780ce60	1287
7a67c85e85bb2ce8582c35f2203ad736	1160
ea8482cd71df3c1969d7b9473ff13abc	1146
4869f7a5dfa277a7dca6462dcf3b52b2	1132
3d871de0142ce09b7081e2b9d1733cb1	1080

Observation

The top seller processed 1,854 orders, while the remaining top sellers processed between 1,080 and 1,806 orders.

Key Insights

- A small group of sellers handles a large number of orders, making them key contributors to the platform's order fulfilment.

Question 26 : Which seller has the best average rating?

Output :

seller_id	average_rating
003554e2dce176b5555353e4f3555ac8	5.00

Observation

The top-rated seller achieved a perfect average rating of 5.00.

Key Insights

- Consistently high customer ratings indicate excellent service quality and can help build customer trust and satisfaction.

Question 27 : Which seller has the fastest delivery performance?

Output :

seller_id	avg_delivery_hours
7ff588a03c2aeae4fbd23f9ae64b760d	11.64

Observation

The fastest-performing seller recorded an average delivery time of 11.64 hours.

Key Insights

- Faster delivery performance can improve customer satisfaction and provides a benchmark for evaluating seller logistics efficiency.

Question 28 : Which seller states generate the highest revenue?

Output :

seller_state	total_revenue
SP	10235883.88
PR	1458900.73
MG	1224159.8
RJ	937814.12
SC	738973.13
RS	435802.63
BA	305262.24
DF	116243.54
PE	103886.31
GO	78964.71

Observation

São Paulo (SP) generated the highest seller revenue (10,235,883.88), followed by Paraná (PR) and Minas Gerais (MG).

Key Insights

- Sellers from SP contribute the largest share of revenue, making it the strongest-performing seller state on the platform.

Question 29: What is the average delivery time?

Output :

avg_delivery_days
12.54

Observation

The average delivery time across all orders is 12.54 days.

Key Insights

- An average delivery time of around 13 days provides a benchmark for evaluating logistics performance and identifying opportunities to improve delivery efficiency.

Question 30 : What percentage of orders are delivered late?

Output :

late_order_percentage
89.15

Observation

Approximately 89.15% of orders were delivered later than the estimated delivery date.

Key Insights

- The high percentage of late deliveries indicates a significant opportunity to improve logistics and delivery performance.

Question 31 : Which states have the longest delivery time?

Output :

customer_state	avg_delivery_days
RR	29.36
AP	27.17
AM	26.40
AL	24.52
PA	23.75

Observation

Roraima (RR) recorded the longest average delivery time (29.36 days), followed by Amapá (AP) and Amazonas (AM).

Key Insights

- States in the northern region experience the longest delivery times, indicating potential logistics and transportation challenges.

Question 32 : Does delivery delay affect review score?

Output :

delivery_status	avg_review_score	total_orders
On Time	4.29	89949
Late	2.27	6410
Not Delivered	1.76	2865

Observation

Orders delivered on time received the highest average review score (4.29), while late and not delivered orders received average scores of 2.27 and 1.76, respectively.

Key Insights

- Delivery performance has a strong impact on customer satisfaction, with delayed or undelivered orders resulting in significantly lower review ratings.

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5. Payment & Review Analysis

Purpose: Analyse payment behaviour and customer satisfaction.

Question 33 : Which payment method is most popular?

Output :

payment_type	total_orders
credit_card	76505

Observation

Credit Card is the most popular payment method, accounting for 76,505 orders.

Key Insights

- Credit card is the preferred payment option among customers, highlighting its importance for payment processing and customer convenience.

Question 34 : Which payment method generates the highest revenue?

Output :

payment_type	total_revenue
credit_card	12542084.19

Observation

Credit Card payments generated the highest revenue, totalling 12,542,084.19.

Key Insights

- Credit card is the primary revenue-generating payment method, making it the most valuable payment option for the business.

Question 35 : What is the average payment value?

Output :

average_payment_value
154.1

Observation

The average payment value is 154.10.

Key Insights

- Customers spend an average of 154.10 per payment transaction, providing a useful benchmark for payment and spending analysis.

Question 36 : How do installment payments affect order value?

Output :

payment_installments	avg_order_value	total_orders
0	94.32	2
1	112.42	49060
2	127.23	12389
3	142.54	10443
4	163.98	7088
5	183.47	5234
6	209.85	3916
7	187.67	1623
8	307.74	4253
9	203.44	644
10	415.09	5315
11	124.93	23
12	321.68	133
13	150.46	16
14	167.96	15
15	445.55	74
16	292.69	5
17	174.6	8
18	486.48	27
20	615.8	17
21	243.7	3
22	228.71	1
23	236.48	1
24	610.05	18

Observation

Average order value generally increases with the number of payment installments. Most orders are paid in 1 installment (49,060 orders), while higher installment plans have fewer orders but higher average order values.

Key Insights

- Customers tend to use more installments for higher-value purchases, while single-installment payments remain the most common payment choice.

Question 37 : What is the average review score?

Output :

average_review_score
4.09

Observation

- The average customer review score is 4.09 out of 5, indicating that most orders received positive ratings.

Key Insight

- Overall customer satisfaction is high, suggesting a generally positive shopping experience, though some orders with lower ratings may be linked to issues such as delivery delays or product/service quality.

Question 38 : Which product categories receive the best ratings?

Output :

product_category_name	average_rating	total_reviews
livros_interesse_geral	4.45	549
construcao_ferramentas_ferramentas	4.44	99
livros_importados	4.40	60
livros_tecnicos	4.37	266
malas_acessorios	4.32	1088
alimentos_bebidas	4.32	279
portateis_casa_forno_e_cafe	4.30	76
fashion_calcados	4.23	261
alimentos	4.22	495
cine_foto	4.21	73

Observation

- livros_interesse_geral has the highest average rating (4.45) among categories with a substantial number of reviews (549).
- Most of the top-rated product categories have an average rating above 4.2, indicating consistently positive customer feedback.

Key Insight

- Book-related categories consistently achieve the highest customer satisfaction, while the large review counts for categories like malas_acessorios and alimentos_bebidas indicate that high ratings are maintained even with significant sales volume, reflecting strong product quality and customer experience.

Question 39 : Which sellers receive the best ratings?

Output :

seller_id	average_rating	total_reviews
d9bd94811c3338dceb4181f3dbc0c73e	4.82	54
d13e50eaa47b4cbe9eb81465865d8cfc	4.81	67
d566c37fa119d5e66c4e9052e83ee4ea	4.72	64
376a891762bbdecbc02b4b6adec3fdda	4.67	57
080199a181c46c657dc5aa235411be3b	4.61	78
5b925e1d006e9476d738aa200751b73b	4.61	63
116ccb1a1604bc88e4d234a8c23f33de	4.60	61
e882b2a25a10b9c057cc49695f222c19	4.60	57
1bb2bdb95f4841f1bba2c0d2cd83d3c9	4.59	80
289cdb325fb7e7f891c38608bf9e0962	4.58	110

Observation

- The top-performing sellers have average ratings between 4.58 and 4.82, with review counts ranging from 54 to 110, indicating consistently positive customer feedback.
- The highest-rated seller achieved an average rating of 4.82 based on 54 reviews.

Key Insight

- Several sellers consistently maintain excellent customer ratings, suggesting that reliable service and product quality contribute to higher customer satisfaction. These sellers can serve as benchmarks for improving overall marketplace performance.

Question 40 : Does review length correlate with review score?

Output :

review_score	avg_review_length	total_reviews
1	99.94	8745
2	96.85	2145
3	83.33	3557
4	61.73	5976
5	51.84	20554

Observation

- 1-star reviews have the longest average review message length (99.94 characters), while 5-star reviews have the shortest (51.84 characters).
- 5-star reviews are the most common (20,554 reviews), whereas 2-star reviews are the least frequent (2,145 reviews).

Key Insight

- Customers who have negative experiences tend to provide more detailed feedback, while satisfied customers typically leave shorter reviews. The high volume of 5-star reviews also indicates strong overall customer satisfaction**.**

6. Advance SQL & Time Analysis

Purpose: Some more relevant analysis.

Question 41 : Which months generate the highest sales?

Output :

month	total_sales
May	1746900.97
August	1696821.64
July	1658923.67
March	1609515.72
April	1578573.51
June	1535156.88
February	1284371.35
January	1253492.22
November	1194882.8
December	878421.1
October	839358.03
September	732454.23

Observation

- May recorded the highest total sales (1,746,900.97), followed by August and July.
- September generated the lowest sales (372,454.23), with October, December, and November also showing relatively low sales compared to other months.

Key Insight

- Sales exhibit clear seasonal fluctuations, with peak performance occurring in the May–August period, while the September–December period experiences a noticeable decline in sales.

Question 42 : Which hours have the highest order volume?

Output :

hour	total_orders
16	6675
11	6578
14	6569
13	6518
15	6454
21	6217
20	6193
10	6177
17	6150
12	5995

Observation

- The highest order volume occurs during the afternoon and early evening, with 4 PM (16:00) recording the most orders (6,675), followed closely by 11 AM, 2 PM, and 1 PM.
- Most peak ordering activity is concentrated between 10 AM and 9 PM.

Key Insight

- Customer purchasing activity is strongest during the daytime and early evening, indicating these hours are the peak demand period for the platform.

Question 43 : Which weekdays have the highest order volume?

Output :

weekday	total_orders
Monday	16196
Tuesday	15963
Wednesday	15552
Thursday	14761
Friday	14122
Sunday	11960
Saturday	10887

Observation

- Monday recorded the highest order volume (16,196 orders), followed by Tuesday and Wednesday.
- Saturday had the lowest order volume (10,887 orders), with Sunday also experiencing comparatively lower demand.

Key Insight

- Customer purchasing activity is strongest on weekdays, particularly at the beginning of the week, while weekends see a noticeable decline in order volume.

Question 44 : Which regions generate the highest revenue?

Output :

region	total_revenue
Southeast	10340831.46
South	2325141.35
Northeast	1898479.44
Central-West	1029797.52
North	414622.35

Observation

- The Southeast region generated the highest revenue (10,340,831.46), contributing significantly more than any other region.
- The North region recorded the lowest revenue (414,622.35), while the South, Northeast, and Central-West generated comparatively moderate revenue**.**

Key Insight

- Revenue is highly concentrated in the Southeast region, making it the primary revenue-driving market. In contrast, the North contributes the least, indicating a significant regional imbalance in sales performance.

Question 45 : Which regions have the highest Average Order Value?

Output :

region	average_order_value
North	224
Northeast	202.09
Central-West	178.1
South	164.34
Southeast	151.48

Observation

- The North region has the highest Average Order Value (AOV) at 224.00, followed by the Northeast (202.09) and Central-West (178.10).
- The Southeast, despite being the highest revenue-generating region, has the lowest AOV (151.48).

Key Insight

- Regions with the highest revenue do not necessarily have the highest order values. The North has fewer but higher-value orders on average, while the Southeast's strong revenue is likely driven by a larger volume of orders rather than higher-value purchases.

Question 46 : Which products generate high revenue but low sales volume?

Output :

product_id	units_sold	total_revenue
bb50f2e236e5eea0100680137654686c	195	63885
6cdd53843498f92890544667809f1595	156	54730.2
d6160fb7873f184099d9bc95e30376af	35	48899.34
d1c427060a0f73f6b889a5c7c61f2ac4	343	47214.51
99a4788cb24856965c36a24e339b6058	488	43025.56
3dd2a17168ec895c781a9191c1e95ad7	274	41082.6
25c38557cf793876c5abdd5931f922db	38	38907.32
5f504b3a1c75b73d6151be81eb05bdc9	63	37733.9
53b36df67ebb7c41585e8d54d6772e08	323	37683.42
aca2eb7d00ea1a7b8ebd4e68314663af	527	37608.9

Observation

- Several products generate high revenue despite relatively low sales volume. For example, product d6160fb7873f184099d9bc95e30376af sold only 35 units but generated 48,899.34 in revenue, while product 25c38557cf793876c5abdd5931f922db sold 38 units and generated 38,907.32.
- In contrast, some products sold hundreds of units but generated lower or comparable revenue, indicating a lower revenue per unit.

Key Insight

- High-priced products can contribute significantly to revenue even with low sales volume, making them valuable contributors to overall business performance. These products represent premium offerings that have a strong impact on revenue despite limited demand.

Question 47 : Which weight category contributes the most revenue?

Output :

weight_category	total_revenue
Heavy	5477214.01
Medium	4740227.38
Light	3374202.31

Observation

- Heavy-weight products generated the highest revenue (5,477,214.01), followed by Medium-weight products (4,740,227.38).
- Light-weight products contributed the lowest revenue (3,374,202.31).

Key Insight

- Heavier products contribute more to overall revenue, suggesting they are either higher-priced or purchased in sufficient quantities to generate greater sales value compared to lighter products.

Question 48 : Rank customers by revenue using RANK().

Output :

customer_rank	customer_id	total_revenue
1	1617b1357756262bfa56ab541c47bc16	13664.08
2	ec5b2ba62e574342386871631fafd3fc	7274.88
3	c6e2731c5b391845f6800c97401a43a9	6929.31
4	f48d464a0baaea338cb25f816991ab1f	6922.21
5	3fd6777bbce08a352fddd04e4a7cc8f6	6726.66
6	05455dfa7cd02f13d132aa7a6a9729c6	6081.54
7	df55c14d1476a9a3467f131269c2477f	4950.34
8	e0a2412720e9ea4f26c1ac985f6a7358	4809.44
9	24bbf5fd2f2e1b359ee7de94defc4a15	4764.34
10	3d979689f636322c62418b6346b1c6d2	4681.78

Observation

- The highest-spending customer generated 13,664.08 in total revenue, while the 10th-ranked customer contributed 4,681.78.
- The top 10 customers each generated more than 4,600 in revenue, indicating a significant contribution from a relatively small group of customers.

Key Insight

- A small segment of high-value customers contributes disproportionately to overall revenue, highlighting the importance of retaining and engaging these customers through targeted loyalty and personalization strategies.

Question 49 : Calculate month-over-month revenue growth using LAG().

Output :

month	revenue	previous_month_revenue	month_growth_percent
April	1578573.51	NULL	NULL
August	1696821.64	1578573.51	7.49
December	878421.1	1696821.64	-48.23
February	1284371.35	878421.1	46.21
January	1253492.22	1284371.35	-2.4
July	1658923.67	1253492.22	32.34
June	1535156.88	1658923.67	-7.46
March	1609515.72	1535156.88	4.84
May	1746900.97	1609515.72	8.54
November	1194882.8	1746900.97	-31.6
October	839358.03	1194882.8	-29.75
September	732454.23	839358.03	-12.74

Observation

- Revenue experienced the highest month-over-month growth in February (+46.21%), followed by July (+32.34%) and May (+8.54%).
- The largest declines occurred in December (-48.23%), November (-31.60%), and October (-29.75%), indicating substantial drops in revenue during these months.

Key Insight

- Revenue shows noticeable month-to-month fluctuations, with periods of strong growth followed by sharp declines. This indicates the presence of seasonal demand patterns and varying customer purchasing behaviour throughout the year.

Question 50 : Find the cumulative revenue over time using window functions.

Output :

month	revenue	cumulative_revenue
April	1578573.51	1578573.51
August	1696821.64	3275395.15
December	878421.1	4153816.25
February	1284371.35	5438187.6
January	1253492.22	6691679.819999999
July	1658923.67	8350603.489999999
June	1535156.88	9885760.37
March	1609515.72	11495276.09
May	1746900.97	13242177.06
November	1194882.8	14437059.860000001
October	839358.03	15276417.89
September	732454.23	16008872.120000001

Observation

- Cumulative revenue increased consistently over the analyzed months, reaching a final total of 15,008,872.12.
- The most significant increases in cumulative revenue occurred during the high-sales months, particularly May, July, and August.

Key Insight

- The steady growth in cumulative revenue indicates sustained business performance over time, with peak sales months contributing the largest share to overall revenue growth.